HH-I-525A
October 4, 1968
SUPERSEDING
Fed. Spec. HH-I-525
April 11, 1960

FEDERAL SPECIFICATION

INSULATION BOARD, THERMAL, CORK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE

1.1 This specification covers cork insulation board for thermal insulation.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Steel Strapping, Flat.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

FSC 5640

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American Trucking Associations, Inc.

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P St., N.W., Washington, D.C. 20036.)

Uniform Classification Committee

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The board shall consist of pure granulated cork bonded by the normal resins of cork.

3.2 Fabrication. The board shall be manufactured from pure granulated cork without the addition of any material foreign to cork.

3.3 Dimensions and tolerances. The board shall be true to form and dimensions, the corners shall be square, and the sides and ends shall be parallel. The board shall conform to the dimensions and tolerances of Section 4, ASTM C352, for the width, length, and thickness specified (see 6.2).

3.4 Thermal conductivity. The maximum thermal conductivity, when tested in accordance with 4.5.1, shall not exceed 0.28 B.t.u. inches per hour sq. ft. degree F. at a mean temperature of 75° F.

3.5 Water absorption. The volume of water absorbed from the increase in weight shall average 5 percent by volume after conditioning when tested in accordance with 4.5.2.

3.6 Density. The board shall weigh not less than 6 and not more than 9 pounds per cubic foot when tested in accordance with 4.5.3.

3.7 Flexural strength. The board shall have a flexural strength of not less than 15 pounds per square inch, when tested in accordance with 4.5.4.

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4.3 Visual and dimensional examination. Each of the sample boards selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any board in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective boards in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Lot acceptance tests. Each of the sample boards selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.5 to verify compliance with this specification. If any board fails any one of the tests of 4.5, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 Thermal conductivity. When specified (see 6.2), the thermal conductivity shall be determined in accordance with the provisions of ASTM C-177.

4.5.2 Water absorption. The water absorption shall be determined in accordance with the provisions of ASTM C209 except that the specimen shall be submerged for 24 hours.

4.5.3 Density. The density shall be determined in accordance with the provisions of ASTM C303.

4.5.4 Flexural strength. The flexural strength shall be determined in accordance with the provisions of paragraph 8c of ASTM C352.

4.5.5 Boiling test. Two specimens, measuring 1 by 5 inches by the thickness of the board, shall be floated on boiling water in a loosely covered vessel for 3 hours. The specimens shall then be removed and examined for evidence of disintegration.

4.6 Inspection of preparation for delivery. For the purpose of inspection a completed pack, prepared for shipment, shall be considered a unit of product. Sampling for examination shall be in accordance with MIL-STD-105, and shall be examined for the defects listed in 4.6.1, using an AQL of 2.5 defects per hundred units.

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6. NOTES

6.1 Intended use. The board covered by this specification is intended for use as a thermal insulation.

6.2 Ordering data. Purchasers may exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Width, length, and thickness required (see 3.3).

c. When fungus proofing is required (see 3.8).

d. Finish required (see 3.10 and 6.3).

e. When thermal conductivity test should be conducted (see 4.5.1).

f. Level of packing required (see 5.1).

g. Marking (see 5.2).

6.3 When the exposed surface of the board is to be painted, procurement documents should provide that one side thereof shall be sanded to a smooth finish (see 3.10).

MILITARY INTEREST
Custodians:
Army - ME
Navy - YD
Air Force - 84

Review activities:
Army - CE
Navy - SH

User activity:
Navy - MC

Preparing activity:
Navy - YD

CIVIL AGENCY INTEREST
GSA - FSS
AGR
HEW
JUS
COM

7

Order for this publication are to be placed with Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents reference herein.

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